Chaitanya Nagesh

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Education

M.S in Computer Science | George Mason University
Concentration: Machine Learning
Coursework: AI, Data Mining, Machine Learning, Deep Learning

B.S in Computer Science | George Mason University
Minor: Software Engineering
Coursework: Data Structures/Analysis of Algorithms, Systems Programming, Software Development, etc

Technical Skills

Programming Languages: Python, Java, C
AI/ML: PyTorch, TensorFlow, Amazon Bedrock, AWS SageMaker, Google Vertex AI
Full Stack: JavaScript, Node.js, SQL, MongoDB, API Development, Git
Cloud Tools: AWS Lambda, S3, API Gateway, DynamoDB, CI/CD, AWS Networking services

Certifications: AWS Cloud Practitioner, AWS AI Practitioner, AWS ML Engineer Associate (in progress), CodePath Technical Interview Preparation

Experience

AI/ML Researcher | Algoverse Feb 2026 - Jul 2026

- Conducted a mechanistic interpretability study on latent chain-of-thought (CoT) models, analyzing internal reasoning behavior using PyTorch, Hugging Face Transformers, and TransformerLens, with techniques such as activation patching and attention analysis.
- Designed and evaluated controlled experiments on arithmetic reasoning tasks (e.g., GSM8K-style), leveraging Python, NumPy, and Matplotlib for data analysis and visualization to assess the stability and structure of latent reasoning representations.

Projects

WasteWise (AI-Powered Food Waste Reduction Web App) | In Development

Tools: Python, FastAPI, React, PostgreSQL, AWS

- Developing a full-stack web application that helps users reduce household food waste through digital pantry management and expiration tracking.
- Building a recommendation system to prioritize recipes based on available ingredients and approaching expiration dates.
- Designing a cloud-based architecture using AWS services for authentication, storage, deployment, and notifications.
- Creating an analytics dashboard to track food saved, estimated cost savings, and environmental impact.

CPU Scheduler | CS 367- Systems Programming

- Designed and developed a CPU Scheduler (written in C) that creates and manages specific tasks and their embedded features.
- Utilized my knowledge of data structures by incorporating linked lists and queues.
- Implemented a thorough testing strategy to find bugs and manage edge cases.

Data Compression Using Huffman Coding | Data Structures and Algorithms

- Developed a program that analyzes a text file to determine the frequency of each character.
- Utilized a priority queue to construct a binary tree where the least frequent characters have higher depth, and more frequent characters have shorter paths.